

Chapter 3.2 Principles of Simple Regression Analysis

§ 3.2.1 Pearson's correlation analysis

§ 3.2.2 Simple linear regression analysis

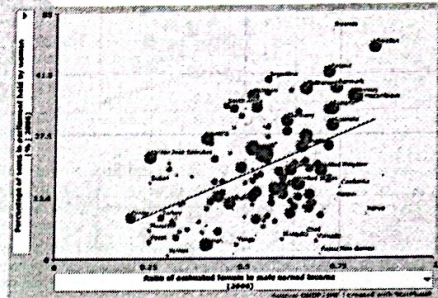
§ 3.2.3 Regression Model Through the Origin

§ 3.2.4 Regression analysis with both independent and dependent variables as random variables

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§ 3.2.1 Pearson's correlation analysis

- ◆ Co-variation or co-relation between two variables
- ◆ These variables change together
- ◆ Usually scale (interval or ratio) variables



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协方差 Covariance

$$\sigma_{XY} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

- $\sigma_{XY} > 0$ X and Y are positively correlated
- $\sigma_{XY} < 0$ X and Y are inversely correlated
- $\sigma_{XY} = 0$ X and Y are independent

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Correlation coefficients

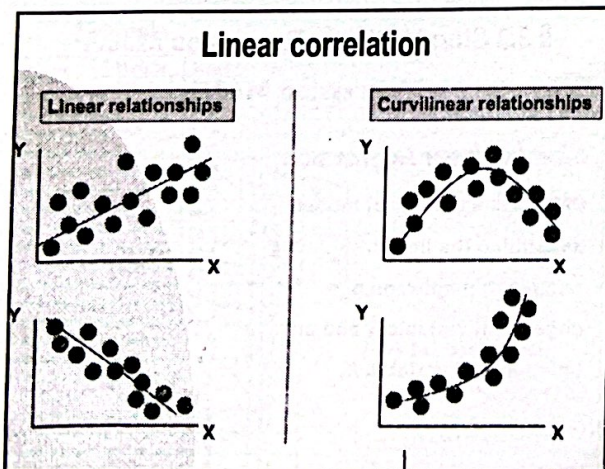
Pearson's Correlation Coefficient between variables x and y is a standardized covariance (unitless):

$$\rho = \frac{\sigma_{XY}}{\sqrt{\sigma_X^2} \sqrt{\sigma_Y^2}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

- Measures the relative strength of the linear relationship between two variables
- Ranges between -1 and 1
- The closer to -1, the stronger the negative linear relationship
- The closer to 1, the stronger the positive linear relationship
- The closer to 0, the weaker any linear relationship

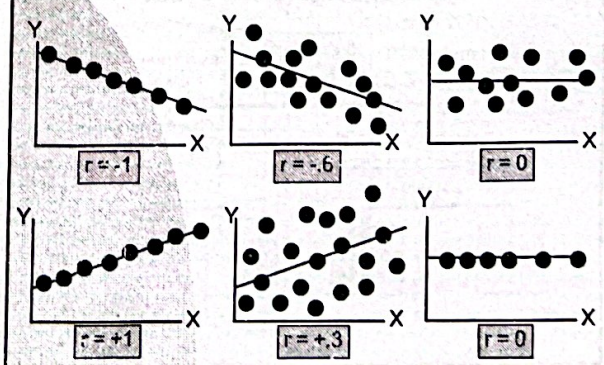
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Linear correlation



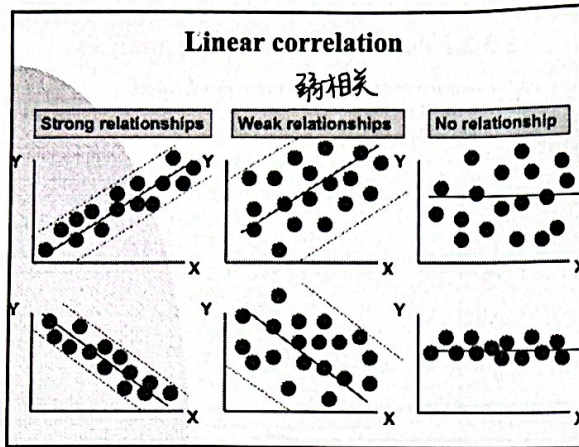
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Scatter plots of data with various correlation coefficients



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Pearson 相关系数为 0.
因为数据不是 Linear fit



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Simpler calculation formula...

$$\hat{\rho} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

Numerator of covariance
 $\hat{\rho} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$
Numerators of variance

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显著性检验 z转化

Test significance of correlation coefficient

z transformation of correlation coefficient $\hat{\rho}$

$$z' = \frac{1}{2} \ln \left(\frac{1 + \hat{\rho}}{1 - \hat{\rho}} \right)$$

When the sample size is large ($n \geq 25$), the distribution of z' is asymptotically normal and has a mean and variance:

$$E(z') = \frac{1}{2} \ln \left(\frac{1 + \rho}{1 - \rho} \right)$$

$$\sigma^2(z') = \frac{1}{n-3}$$

渐近正态分布

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显著水平

Confidence interval $(1 - \alpha)$ of the correlation coefficient $\hat{\rho}$

置信区间

$$\frac{e^{2u_1} - 1}{e^{2u_1} + 1} \leq \rho \leq \frac{e^{2u_2} - 1}{e^{2u_2} + 1}$$

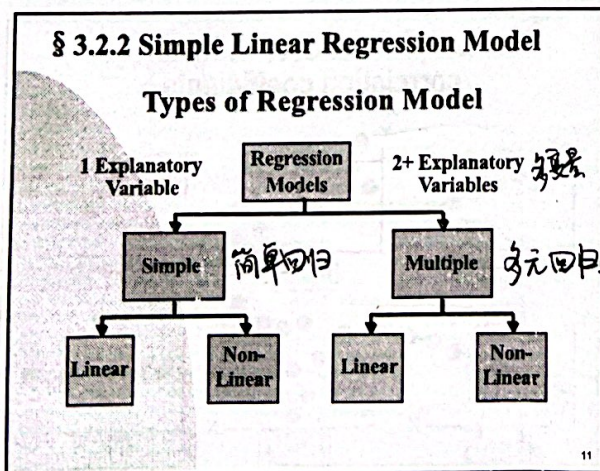
where

$$u_1 = \frac{1}{2} \ln \left(\frac{1 + r}{1 - r} \right) - z(\alpha/2) \sqrt{\frac{1}{n-3}}$$

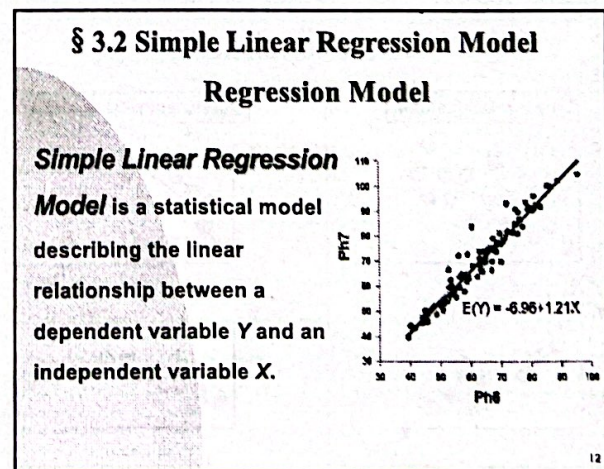
$$u_2 = \frac{1}{2} \ln \left(\frac{1 + r}{1 - r} \right) + z(\alpha/2) \sqrt{\frac{1}{n-3}}$$

正态分布

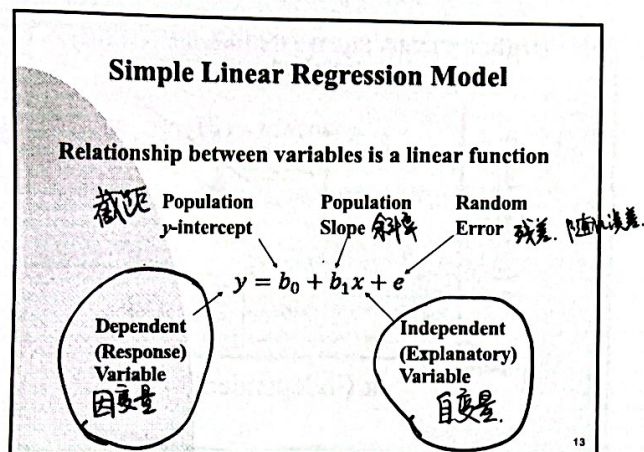
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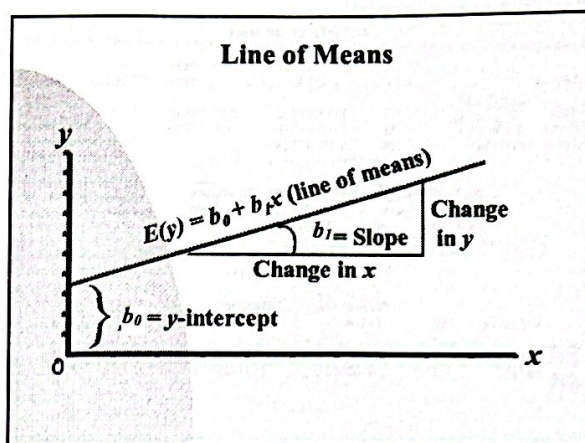
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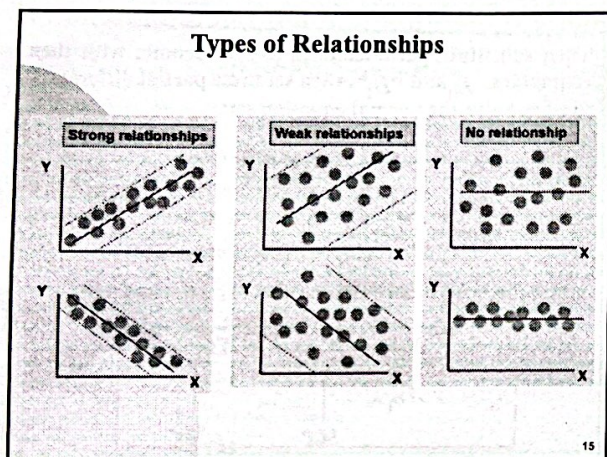


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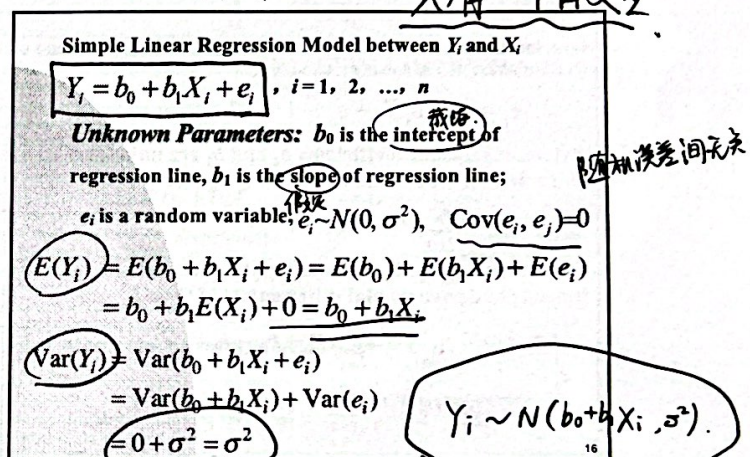


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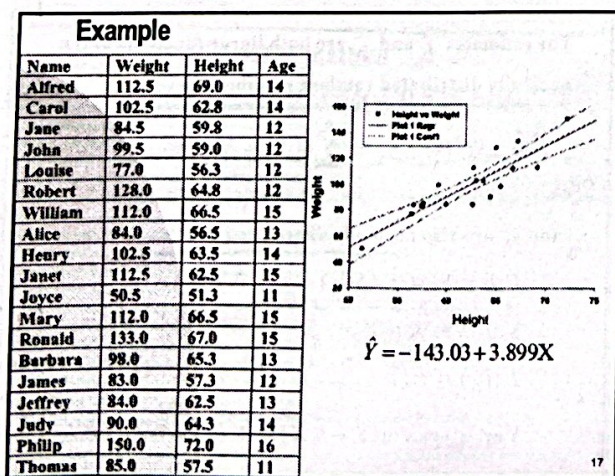
b_0, b_1 参数
 X_i 自变量
 e_i 随机变量
只有一个自变量



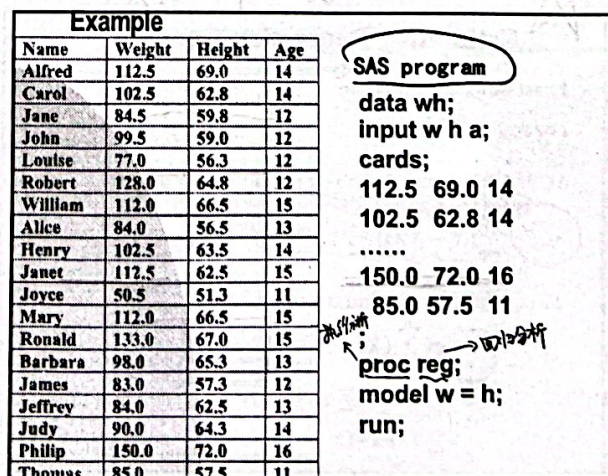
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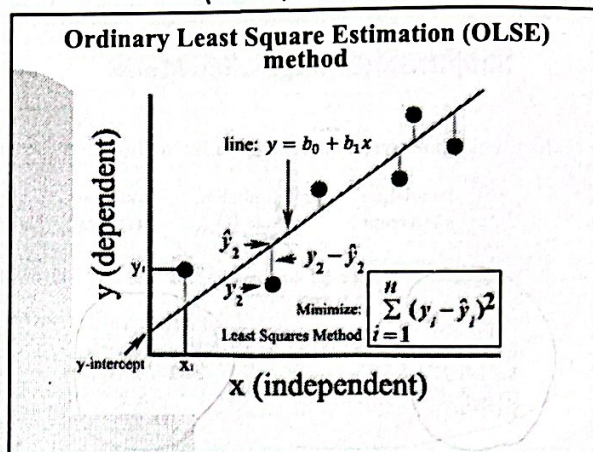
最小二乘法

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	7193.24912	7193.24912	57.08	<.0001
Error	17	2142.48772	126.02869		
Corrected Total	18	9335.73684			

Root MSE	11.22625	R-Square	0.7705
Dependent Mean	100.02632	Adj R-Sq	0.7570
Coeff Var	11.22330		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-143.02632	32.27459	-4.43	0.0004
Height	1	3.89303	0.51609	7.55	<.0001

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Now, let us use a simple linear model to demonstrate the procedure of OLSE method. If Y_i is linearly related to X_i

$$Y_i = b_0 + b_1 X_i + e_i, \quad i = 1, 2, \dots, n$$

where, regression coefficients b_0 and b_1 are unknown parameters. The formula of sum of square Q is

$$Q = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$$

For estimating b_0, b_1 and making the Q reach minimum, it needs to derive partial differentials of Q on b_0 and b_1

$$\begin{cases} \partial Q / \partial b_0 = -2 \sum_{i=1}^n (Y_i - b_0 - b_1 X_i) \\ \partial Q / \partial b_1 = -2 \sum_{i=1}^n X_i (Y_i - b_0 - b_1 X_i) \end{cases}$$

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Then substitute parameters in above formula with their estimators $\hat{b}_0$ and $\hat{b}_1$, then set these partial differential to zero. Solve the normal equation set:

$$\begin{cases} -2 \sum_{i=1}^n (Y_i - \hat{b}_0 - \hat{b}_1 X_i) = 0 \\ -2 \sum_{i=1}^n X_i (Y_i - \hat{b}_0 - \hat{b}_1 X_i) = 0 \end{cases}$$

obtain the least square estimators of two parameters

$$\begin{aligned} \hat{b}_1 &= (\sum_{i=1}^n X_i Y_i - n \bar{X} \bar{Y}) / (\sum_{i=1}^n X_i^2 - n \bar{X}^2) \\ \hat{b}_0 &= \bar{Y} - \hat{b}_1 \bar{X} \end{aligned}$$

SCP SS_X

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$$\hat{b}_1 = \frac{SCP}{SS_X} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}$$

$\hat{b}_0$ 和 $\hat{b}_1$ 也是正态分布.

★ Estimating Regression Equation

Least Square Estimation $\hat{b}_1 = SCP / SS_X$

Formula of parameter $\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}$

SCP is the sum of cross products of Y and X

$$SCP = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n \bar{X} \bar{Y} = \sum_{i=1}^n (X_i - \bar{X}) Y_i$$

SS_X is the sum of squares of X

$$SS_X = \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n \bar{X}^2$$

$$SSR = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (b_0 + \hat{b}_1 X_i - \bar{Y})^2$$

$$= \sum_{i=1}^n (\bar{Y} - \hat{b}_1 \bar{X} + \hat{b}_1 X_i - \bar{Y})^2 = \hat{b}_1^2 SS_X$$

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The estimates $\hat{b}_0$ and $\hat{b}_1$ are both linear functions of the normally distributed random variable Y_i ,

$$\hat{b}_1 = SCP / SS_X = \sum_{i=1}^n [(X_i - \bar{X}) / SS_X] Y_i$$

$\hat{b}_0$ and $\hat{b}_1$ are also normally distributed random variables.

$$E(\hat{b}_1) = E(SCP / SS_X) = b_1 \quad \text{无偏}$$

$$Var(\hat{b}_1) = Var(SCP / SS_X) = \sigma^2 / SS_X$$

$$E(\hat{b}_0) = E(\bar{Y} - \hat{b}_1 \bar{X}) = b_0 \quad \text{无偏}$$

$$Var(\hat{b}_0) = Var(\bar{Y} - \hat{b}_1 \bar{X}) = \sigma^2 (1/n + \bar{X}^2 / SS_X)$$

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$$\sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right)$$

$$\hat{b}_1 \sim N(b_1, \frac{\sigma^2}{SS_X})$$

$$\hat{b}_0 \sim N(b_0, \sigma^2 (\frac{1}{n} + \frac{\bar{X}^2}{SS_X}))$$

$$SCP = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n (X_i - \bar{X}) Y_i - \sum_{i=1}^n (X_i - \bar{X}) \bar{Y}$$

$$= \sum X_i Y_i - n \bar{X} \bar{Y} - n \bar{X} \bar{Y} + n \bar{X} \bar{Y}$$

$$= \sum X_i Y_i - n \bar{X} \bar{Y}$$

回归系数无偏估计的证明

$$\begin{aligned}
 E(\hat{b}_1) &= E(SCP/SS_X) \\
 &= (1/SS_X)E(SCP) \\
 &= (1/SS_X)E\left(\sum_{i=1}^n (X_i - \bar{X})Y_i\right) \\
 &= (1/SS_X)E\left(\sum_{i=1}^n (X_i - \bar{X})(b_0 + b_1 X_i)\right) \\
 &= (1/SS_X)\left(\sum_{i=1}^n (X_i - \bar{X})X_i b_1\right) \\
 &= (1/SS_X)SS_X b_1 \\
 &= b_1
 \end{aligned}$$

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回归系数估计值的方差

$$\begin{aligned}
 \text{Var}(\hat{b}_1) &= \text{Var}(SCP/SS_X) \\
 &= (1/SS_X)^2 \text{Var}(SCP) \\
 &= (1/SS_X)^2 \sum_{i=1}^n (X_i - \bar{X})^2 \text{Var}(Y_i) \\
 &= (1/SS_X) \sigma^2 \\
 &= \sigma^2 / SS_X \\
 &= \sigma^2(\hat{b}_1)
 \end{aligned}$$

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回归截距的无偏估计和方差

$$\begin{aligned}
 E(\hat{b}_0) &= E(\bar{Y} - \hat{b}_1 \bar{X}) \\
 &= E(\bar{Y}) - E(\hat{b}_1 \bar{X}) \\
 &= b_0 + b_1 \bar{X} - b_1 \bar{X} \\
 &= b_0 \\
 \text{Var}(\hat{b}_0) &= \text{Var}(\bar{Y} - \hat{b}_1 \bar{X}) \\
 &= \text{Var}(\bar{Y}) + \text{Var}(\hat{b}_1 \bar{X}) - 2\text{Cov}(\bar{Y}, \hat{b}_1 \bar{X}) \\
 &= \sigma^2/n + \sigma^2 \bar{X}^2 / SS_X \\
 &\quad - \frac{2\bar{X}}{n SS_X} [\sum_{i=1}^n (X_i - \bar{X}) \text{Var}(Y_i) + \sum_{i=1}^n (X_i - \bar{X}) \text{Cov}(Y_i, Y_j)] \\
 &= \sigma^2 (1/n + \bar{X}^2 / SS_X) \\
 &= \sigma^2(\hat{b}_0)
 \end{aligned}$$

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最大似然法

Maximum Likelihood Estimation (MLE) method

Recall:

$$Y_i = b_0 + b_1 X_i + e_i, \quad i = 1, 2, \dots, n$$

 Y_i is normally distributed:

$$Y_i \sim N(b_0 + b_1 X_i, \sigma^2)$$

The probability density function (pdf) of Y_i :

$$p(y_i | x_i; b_0, b_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (b_0 + b_1 x_i))^2}{2\sigma^2}}$$

概率密度函数

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MLE method

找到一组 para, 使观察到这组数据的概率最大

Given any data set $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, we can now write down the probability density, under the model, of seeing that data:

$$\prod_{i=1}^n p(y_i | x_i; b_0, b_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (b_0 + b_1 x_i))^2}{2\sigma^2}}$$

This is the likelihood, a function of the parameter values.

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MLE method

Log-likelihood:

$$\begin{aligned}
 L(b_0, b_1, \sigma^2) &= \log \prod_{i=1}^n p(y_i | x_i; b_0, b_1, \sigma^2) \\
 &= \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (b_0 + b_1 x_i))^2}{2\sigma^2}} \\
 &= -\frac{n}{2} \log 2\pi - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2
 \end{aligned}$$

Derivative of log-likelihood:

$$\frac{\partial L}{\partial b_1} = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - (b_0 + b_1 x_i)) x_i$$

$$\frac{\partial L}{\partial b_0} = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))$$

$$\frac{\partial L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2$$

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参数估计值

MLE method

MLE of b_0, b_1, σ^2 :

$$\begin{cases} \hat{b}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{SCP}{SS_X} \\ \hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x} \\ \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - (\hat{b}_0 + \hat{b}_1 x_i))^2 \end{cases}$$

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Several Expressions of Simple Regression:

Correct Expressions:

$$Y_i = b_0 + b_1 X_i + e_i$$

$$\hat{Y}_i = \hat{b}_0 + \hat{b}_1 X_i \text{ or } \hat{Y} = \hat{b}_0 + \hat{b}_1 X$$

$$E(\hat{Y}_i) = b_0 + b_1 X_i \text{ or } E(\hat{Y}) = b_0 + b_1 X$$

$$E(Y_i) = b_0 + b_1 X_i$$

Incorrect Expressions:

$$Y_i = b_0 + b_1 X_i + e_i$$

$$\hat{Y}_i = \hat{b}_0 + \hat{b}_1 X_i$$

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Application of Analysis of Variance

方差分析 (analysis of variance, ANOVA):

Partition the total variation to different components due to different causes, and investigate the importance of each components to the total variation.

分析回归模型。
回归的变异和残差的变异进行比较是显著。
将总的变异分解为某些感兴趣变量的变异。

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Formula for decomposition of the total sum of squares (SSTO) of simple linear regression model

Total Sum of Squares $SSTO = \sum_{i=1}^n (Y_i - \bar{Y})^2$

Regression Sum of Squares $SSR = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$

Error Sum of Squares $SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$

Unexplained sum of squares $(y_i - \hat{y}_i)^2$

Explained sum of squares $(\hat{y}_i - \bar{y})^2$

Total sum of squares $(y_i - \bar{y})^2$

可用 $\sum_{i=1}^n [(Y_i - \bar{Y})^2] = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$ 不可用解释的变异 (残差平方和)

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Total Sum of Squares (SSTO) =

Regression Sum of Squares ($SSR = (\hat{b}_1)^2 SS_X$)

+ Residual Sum of Squares ($SSE = SSTO - (\hat{b}_1)^2 SS_X$)

We may also disassemble the degrees of freedom correspondingly

Total degrees of freedom $df_T (n-1)$ =

Degrees of freedom for regression $df_R (1)$

+ Degrees of freedom for residual $df_E (n-2)$

Estimating the mean square (MS)

Total Mean Square $MSTO = SSTO / (n-1)$

Regression Mean Square $MSR = SSR / 1$

$E(MSR) = E(\hat{b}_1^2 SS_X) = \sigma^2 + b_1^2 SS_X$

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因为 $b_1=0$ 时没有显著线性关系

$$MSE = \hat{\sigma}^2$$

Residual Mean Square $MSE = SSE / (n-2)$ 残差的平均

为什么是 σ^2 ? $E(MSE) = \sigma^2$

Under $H_0: b_1 = 0$, F^* should have higher probability F 检验

$F^* = MSR / MSE \sim F_{(1, n-2)}$

If $F^* > F_{(\alpha, 1, n-2)}$, then H_0 is rejected, and there exists a significant linear relationship between the dependent variable Y and the independent variable X

ANOVA table of simple linear regression

Cause	df	SS	MS	Expected MS	F value
Regression	1	SSR	MSR	$\sigma^2 + b_1^2 SS_X$	MSR / MSE
Error	$n-2$	SSE	MSE	σ^2	
Total Sum	$n-1$	SSTO			

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对回归参数进行直接检验

t test

The estimate $\hat{b}_1$ is a linear functions of the normally distributed random variable Y_i ,

$$\hat{b}_1 = SCP / SS_X = \sum_{i=1}^n [(X_i - \bar{X}) / SS_X] Y_i$$

$\hat{b}_1$ is also normally distributed random variable.

$$E(\hat{b}_1) = E(SCP / SS_X) = b_1$$

$$Var(\hat{b}_1) = Var(SCP / SS_X) = \sigma^2 / SS_X$$

↑
估计量

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$$\sigma(\hat{b}_1) = \sqrt{\frac{\sigma^2}{SS_X}}$$

t test

$H_0: b_1 = 0; H_1: b_1 \neq 0$

$$t^* = \frac{\hat{b}_1 - b_1}{\hat{\sigma}(\hat{b}_1)} = \frac{\hat{b}_1 - b_1}{\sqrt{MSE / SS_X}} \quad df = n - 2$$

MSE is an unbiased estimate of σ^2 $E(MSE) = \sigma^2$

Confidence interval (1 - α) of the regression coefficient $\hat{b}_1$

$$\hat{b}_1 - t(\alpha/2; n-2)\hat{\sigma}(\hat{b}_1) \leq b_1 \leq \hat{b}_1 + t(\alpha/2; n-2)\hat{\sigma}(\hat{b}_1)$$

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t test

The estimate $\hat{b}_0$ is a linear functions of the normally distributed random variable Y_i ,

$$\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}$$

$\hat{b}_0$ is also normally distributed random variable.

$$E(\hat{b}_0) = E(\bar{Y} - \hat{b}_1 \bar{X}) = b_0$$

$$Var(\hat{b}_0) = Var(\bar{Y} - \hat{b}_1 \bar{X}) = \sigma^2(1/n + \bar{X}^2 / SS_X)$$

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t test

$H_0: b_0 = 0; H_1: b_0 \neq 0$

$$t^* = \frac{\hat{b}_0 - b_0}{\hat{\sigma}(\hat{b}_0)} \quad df = n - 2$$

$$\hat{\sigma}^2(\hat{b}_0) = MSE(1/n + \bar{X}^2 / SS_X)$$

Confidence interval (1 - α) of the regression intercept $\hat{b}_0$

$$\hat{b}_0 - t(\alpha/2; n-2)\hat{\sigma}(\hat{b}_0) \leq b_0 \leq \hat{b}_0 + t(\alpha/2; n-2)\hat{\sigma}(\hat{b}_0)$$

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simple

决定系数(r^2 , coefficient of determination): Ratio of regression sum of squares and total sum of squares.

$$r^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

It measures the proportion of the total variation of dependent variable Y that is explained by X

In an example $r^2 = SSR/SSTO = 160.0/177.6 = 0.9009$
This indicates that 90% of the variance of the Y variable is due to the X variable.

自变量能解释因变量 90% 的方差

相关性越好

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调整决定系数(r^2 , coefficient of determination): The adjusted r^2 is a modified version of r^2 that has been adjusted for the number of predictors in the model.

$$r_{adj}^2 = 1 - \frac{\frac{1}{n-p-1} SSE}{\frac{1}{n-1} SSTO} = r^2 - (1-r^2) \frac{p}{n-p-1}$$

where n is sample size and p is the number of covariates in the model.

$p \uparrow$ $r_{adj} \downarrow$

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$$\hat{\rho} = \sqrt{\frac{SCP^2}{SS_X SS_Y}}$$

$$r^2 = \frac{SSR}{SSTO}$$

Relationship between Pearson correlation coefficient and determination coefficient

Correlation Coefficient ρ is a parameter measuring the correlation between dependent variable y and independent variable x

$$\begin{aligned}\hat{\rho} &= r = SCP / \sqrt{SS_X SS_Y} \\ &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} \\ &= \sqrt{SCP^2 / [SS_X SS_Y]} \\ &= \sqrt{[SCP / SS_X] [SS_Y / SS_Y]} \\ &= \sqrt{[b_1]^2 SS_X / SS_Y} \\ &= \sqrt{SSR / SSTO} \\ &= \sqrt{r^2}\end{aligned}$$

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$$\rho = \frac{cov(X, Y)}{\sqrt{Var(X) Var(Y)}}$$

§ 3.2.3 Regression Model Through the Origin

The simple linear regression model

through the origin ($b_0 = 0$) is
 $Y_i = bX_i + e_i \sim N(bX_i, \sigma^2) \quad i=1, 2, \dots, n$

Least squares estimator of the regression slope b

$$\hat{b} = (\sum_{i=1}^n X_i Y_i) / (\sum_{i=1}^n X_i^2) \sim N(b, \frac{\sigma^2}{\sum_{i=1}^n X_i^2})$$

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The prediction formula of dependent variables is

$$\hat{Y}_i = \hat{b}X_i$$

The predicted values of dependent variables are normally distributed

$$\hat{Y}_i \sim N(bX_i, \sigma^2 X_i^2 / \sum_{i=1}^n X_i^2)$$

$$SSTO = \sum_{i=1}^n Y_i^2$$

$$SSR = (\hat{b} \sum_{i=1}^n X_i Y_i)$$

$$SSE = \sum_{i=1}^n Y_i^2 - \hat{b} \sum_{i=1}^n X_i Y_i$$

没有Y的校正

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ANOVA table of simple linear regression through the origin

Cause	df	SS	MS	Expected MS	F value
Regression	1	SSR	MSR	$\sigma^2 + b^2 \sum_{i=1}^n X_i^2$	MSR/MSE
Error	$n-1$	SSE	MSE	σ^2	
Total Sum	n	SSTO			

$MSE = SSE/(n-1)$ is an unbiased estimator of σ^2

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Example

Name	Weight	Height	Age
Alfred	112.5	69.0	14
Carol	102.5	62.8	14
Jane	84.5	59.8	12
John	99.5	59.0	12
Lonise	77.0	56.3	12
Robert	128.0	64.8	12
William	112.0	66.5	15
Alice	84.0	56.5	13
Henry	102.5	63.5	14
Janet	112.5	62.5	15
Joyce	50.5	51.3	11
Mary	112.0	66.5	15
Ronald	133.0	67.0	15
Barbara	98.0	65.3	13
James	83.0	57.3	12
Jeffrey	84.0	62.5	13
Judy	90.0	64.3	14
Philip	150.0	72.0	16
Thomas	85.0	57.5	11

SAS program

```
data wh;
input w h a;
cards;
112.5 69.0 14
102.5 62.8 14
84.5 59.8 12
99.5 59.0 12
77.0 56.3 12
128.0 64.8 12
112.0 66.5 15
84.0 56.5 13
102.5 63.5 14
112.5 62.5 15
50.5 51.3 11
112.0 66.5 15
133.0 67.0 15
98.0 65.3 13
83.0 57.3 12
84.0 62.5 13
90.0 64.3 14
150.0 72.0 16
85.0 57.5 11
;
proc reg;
model w = h;
run;
```

We may change it to:
model w = h|noint;

没有校正: 采用原始数据

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Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	7193.24912	7193.24912	57.08	<.0001
Error	17	2142.48772	126.02869		
Corrected Total	18	9335.73684			

方差估计值

Root MSE	11.22625	R-Square	0.7705
Dependent Mean	100.02632	Adj R-Sq	0.7570
Coeff Var	11.22330		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-143.02692	32.27459	-4.43	0.0004
Height	1	3.69903	0.51609	7.55	<.0001

截距

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模型诊断

Simple regression analysis table of weight and height

Source	Regression intercept exists			No regression intercept		
	DF	MS	Pr>F	DF	MS	Pr>F
Model	1	7193.24912	<.0001	1	194818	<.0001
Error	17	126.02869		18	256.52972	

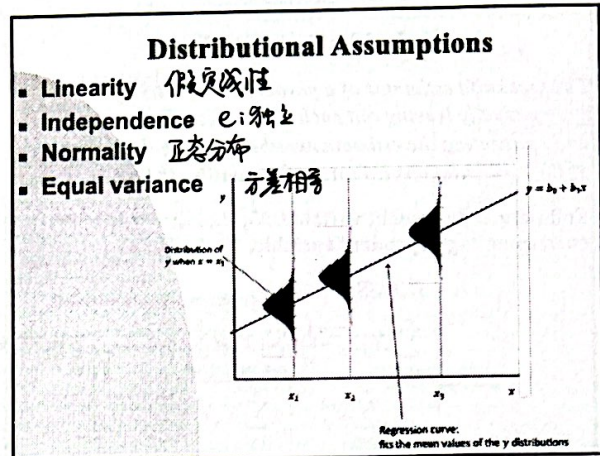
Regression prediction model between weight and height
 包络性 $\hat{Y} = -143.03 + 3.899X$, $R^2 = 0.7705$
 不包络性 $\hat{Y} = 1.619X$, $R^2 = 0.977$

Simple regression analysis table of weight and age

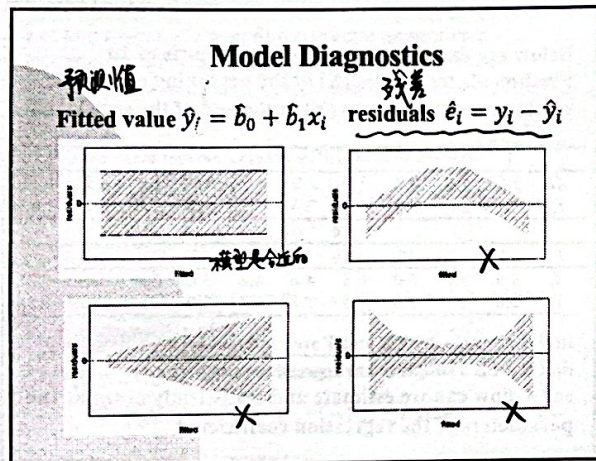
Source	Regression intercept exists			No regression intercept		
	DF	MS	Pr>F	DF	MS	Pr>F
Model	1	5124.49111	0.0003	1	194655	<.0001
Error	17	247.72034		18	265.61833	

Regression prediction model between weight and age
 $\hat{Y} = -50.492 + 11.304X$, $R^2 = 0.5489$
 $\hat{Y} = 7.556X$, $R^2 = 0.975$

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X 为随机变量而不是常数变量.

§ 3.2.4 Regression analysis with both independent and dependent variables as random variables

If X is an independent random variable, its probability distribution does not include regression parameters b_0 , b_1 and σ^2 :

$$X \sim N(\mu_X, \sigma_X^2)$$

Y is also a normal random variable with marginal distribution:

$$Y \sim N(\mu_Y, \sigma_Y^2)$$

There is a correlation between X and Y :

假设 $\rho = \sigma_{XY} / [\sigma_X \sigma_Y] \neq 0$

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Given X , the conditional probability distribution of Y is an independent random variable:

假设 $Y|X=x \sim N(\mu_Y + \sigma_{XY}/\sigma_X^2(x - \mu_X), \sigma_Y^2(1 - \rho^2))$
 $\sim N(b_0 + b_1x, \sigma^2)$

$E(Y|X=x) = \mu_Y + \rho\sigma_Y/\sigma_X(x - \mu_X)$ $\text{Var}(Y|X=x) = \sigma_Y^2(1 - \rho^2)$
 $= \mu_Y + \sigma_{XY}/\sigma_X^2(x - \mu_X)$ $= \sigma_Y^2 - \frac{\sigma_{XY}^2}{\sigma_X^2\sigma_Y^2}\sigma_Y^2$
 $= \mu_Y + b_1(x - \mu_X)$ $= \sigma_Y^2 - b_1^2\sigma_X^2$
 $= (\mu_Y - b_1\mu_X) + b_1x$ $= \sigma^2$
 $= b_0 + b_1x$

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Estimating regression parameters:

$$\hat{b}_1 = \hat{\sigma}_{XY} / \hat{\sigma}_X^2 = \text{SCP} / \text{SS}_X$$

$$\hat{b}_0 = \hat{\mu}_Y - \hat{b}_1\hat{\mu}_X = \bar{Y} - \hat{b}_1\bar{X}$$

$\hat{\sigma}^2 = \hat{\sigma}_Y^2 - \hat{b}_1^2\hat{\sigma}_X^2$
 $= (\text{SS}_Y - \hat{b}_1^2 \text{SS}_X) / (n-1)$
 $< \text{MSE} = (\text{SS}_Y - \hat{b}_1^2 \text{SS}_X) / (n-2)$

比单独方差要小.

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Jackknife resampling

The jackknife estimator of a parameter is found by systematically leaving out each observation from a dataset and calculating the estimate and then finding the average of these calculations (Maurice Quenouille, 1949).

Full data, independent variable, $X_1, \dots, X_n$; corresponding dependent variable, $Y_1, \dots, Y_n$;

$$\hat{b}_1 = SCP/SS_X$$

$$= \frac{\sum_{j=1}^n X_j Y_j - \frac{1}{n} (\sum_{j=1}^n X_j) (\sum_{j=1}^n Y_j)}{\sum_{j=1}^n X_j^2 - \frac{1}{n} (\sum_{j=1}^n X_j)^2}$$

而数据

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每次去除一些数据。

Leave-one-out data, delete (X_i, Y_i) :

$$\hat{b}_{1(i)} = \frac{\sum_{j \neq i} X_j Y_j - \frac{1}{n-1} (\sum_{j \neq i} X_j) (\sum_{j \neq i} Y_j)}{\sum_{j \neq i} X_j^2 - \frac{1}{n-1} (\sum_{j \neq i} X_j)^2}$$

After i -th resampling, the i -th pseudovalue is calculated as:

$$J_i(\hat{b}_1) = n\hat{b}_1 - (n-1)\hat{b}_{1(i)}$$

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Using a total of r pseudovalues, we can get Jackknife estimator of regression coefficient b_1 and its standard error:

$$\hat{J}(\hat{b}_1) = \frac{1}{n} \sum_{i=1}^n J_i(\hat{b}_1)$$

$$SE_J(\hat{b}_1) = \sqrt{\frac{1}{n(n-1)} \sum_{i=1}^n [J_i(\hat{b}_1) - \hat{J}(\hat{b}_1)]^2}$$

$$\left\{ \begin{array}{ll} \frac{J(\hat{b}_1) - b_1}{SE_J(b_1)} \sim t_{df=n-1} & \text{when } n \text{ is small } (n < 30) \\ \frac{J(\hat{b}_1) - b_1}{SE_J(b_1)} \sim N(0,1) & \text{when } n \text{ is large} \end{array} \right.$$

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Below are the observed numerical records of 20 Freshmen's test scores (X) at the beginning of the semester and test scores (Y) at the end of the semester:

i	1	2	3	4	5	6	7	8	9	10
X_i	5.5	4.8	4.7	3.9	4.5	6.2	6.0	5.2	4.7	4.3
Y_i	3.1	2.3	3.0	1.9	2.5	3.7	3.4	2.6	2.8	1.6

i	11	12	13	14	15	16	17	18	19	20
X_i	4.9	5.4	5.0	6.3	4.6	4.3	5.0	5.9	4.1	4.7
Y_i	2.0	2.9	2.3	3.2	1.8	1.4	2.0	3.8	2.2	1.5

In this example, both X and Y are random variables. Assuming that there is a simple linear regression relationship between Y and X , how can we estimate and statistically examine the parameters of the regression coefficient?

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The estimation and standard error of regression parameters are calculated by regular methods

$$\bar{X} = 5, \bar{Y} = 2.5, \sum X_i^2 = 509.12, \sum Y_i^2 = 134.84, \sum X_i Y_i = 257.66,$$

$$SCP = \sum X_i Y_i - n\bar{X}\bar{Y} = 257.66 - 20 \times 5 \times 2.5 = 7.66,$$

$$SS_X = \sum X_i^2 - n\bar{X}^2 = 509.12 - 20 \times 5^2 = 9.12,$$

$$\hat{b}_1 = 0.8399, SE(\hat{b}_1) = 0.1440.$$

$$\hat{b}_0 = -1.6996, SE(\hat{b}_0) = 0.7268$$

无相关性
但实际是有

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Leave-one-out data, delete (X_i, Y_i) :
Estimating b_1 and b_0

i	1	2	3	4	5	6	7	8	9	10
$\hat{b}_{1(i)}$	.829	.839	.866	.888	.865	.808	.832	.841	.859	.813
$\hat{b}_{0(i)}$	-1.66	-1.69	-1.87	-1.96	-1.85	-1.55	-1.66	-1.70	-1.83	-1.55

i	11	12	13	14	15	16	17	18	19	20
$\hat{b}_{1(i)}$	.835	.837	.840	.913	.823	.796	.840	.778	.892	.814
$\hat{b}_{0(i)}$	-1.65	-1.69	-1.69	-2.04	-1.59	-1.45	-1.67	-1.42	-1.99	-1.53

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Jackknife pseudovalues: $J_i(\hat{b}_1)$ and $J_i(\hat{b}_0)$

$$J_i(\hat{b}_1) = n\hat{b}_1 - (n-1)\hat{b}_{1(i)}$$

i	1	2	3	4	5	6	7	8	9	10
$\hat{b}_{1(i)}$	1.043	.853	.340	-.068	.366	1.446	.989	.810	.473	1.348
$\hat{b}_{0(i)}$	-2.53	-1.80	1.56	3.22	1.10	-4.50	-2.38	-1.62	0.69	-4.57

i	11	12	13	14	15	16	17	18	19	20
$\hat{b}_{1(i)}$	.931	.897	.840	-.548	1.165	1.673	.840	2.024	-.153	1.337
$\hat{b}_{0(i)}$	-2.57	-1.92	-1.90	4.75	-3.70	-6.41	-2.20	-7.02	3.77	-4.94

Jackknife estimator of regression coefficients and their standard error:

$$J(\hat{b}_1) = 0.8304, SE_J(\hat{b}_1) = 0.1405$$

$$J(\hat{b}_0) = -1.6483, SE_J(\hat{b}_0) = 0.7313$$

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Orthogonal matrices:

Definition of orthogonality:

Two $n \times 1$ vectors u and v are said to be orthogonal if

$$u'v = v'u = 0$$

A set of $n \times 1$ vectors $\{x_1, x_2, \dots, x_n\}$ is said to be orthonormal if

$$x_i'x_j = 1, x_i'x_j = 0, i \neq j, i, j = 1, 2, \dots, n.$$

Definition of orthogonal matrix:

A $n \times n$ square matrix P is said to be orthogonal if

$$PP' = P'P = I_{nn}$$

$$Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$Q^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$QQ^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q^T = Q^{-1}$$

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For any matrix, we may find a unique generalized inverse, which is called Moore-Penrose generalized inverse. For a matrix, its corresponding Moore-Penrose generalized inverse is denoted by A^+ .

M-P generalized inverse satisfies all following properties:

$$AA^+A = A, A^+AA^+ = A^+, (A^+A)^T = (A^+A), (AA^+)^T = (AA^+)$$

We can get A^+ by $A^+ = Q \begin{bmatrix} D^{-1} & 0 \\ 0 & 0 \end{bmatrix} P^T$

For any matrix X , we have $P_X = X(X^TX)^-X^T$, which is the projection matrix of X .

The properties of projection matrix:

$$P_X X = X, X^T P_X = X^T, P_X^T = P_X, P_X P_X = P_X$$

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Distribution of quadratic form (二次型)

Let the random vector x have a multivariate normal distribution: $x \sim MVN(\mu_X, V_X)$

$$\text{then } x^T A x = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

is the quadratic form of vector x , where A is a symmetric constant matrix.

We often denote the sum of squares by quadratic form in multivariate regression analysis and variance analysis.

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$x^T A y$ is the quadratic form of vector x and y , which is often used to denote the sum of cross products in covariance analysis.

The mean value and variance of quadratic form $x^T A x$:

$$E(x^T A x) = \text{tr}(A V_X) + \mu_X^T A \mu_X$$

$$E(x^T A y) = \text{tr}(A C_{XY}) + \mu_X^T A \mu_Y$$

$$\text{Var}(x^T A x) = 2 \text{tr}(A V_X)^2 + 4 \mu_X^T A V_X A \mu_X$$

$$\text{Cov}(x, x^T A x) = 2 V_X A \mu_X$$

$$\text{Cov}(x^T A x, x^T B x) = 2 \text{tr}(A V_X B V_X) + 4 \mu_X^T A V_X B \mu_X$$

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